

Introduction to Forensic Science

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Lecture Notes

Forensic Sciences

Forensics, simply put, involves applying scientific principles to legal matters. However, the term "law" encompasses various areas where forensics may not apply, such as civil and commercial law. Therefore, a more focused definition would be the use of scientific methods within criminal justice to analyze physical, chemical, and biological evidence from crime scenes and beyond.

Forensic science aids in reconstructing events and determining charges in cases. It can also intersect with archaeology, as both disciplines aim to understand past events. Edmond Locard, a French professor, is considered the pioneer of forensic science, famously stating that "Every contact leaves a trace." Physical evidence is reliable and doesn't lie, but errors in forensic science usually stem from human mistakes in the process.

Forensic sciences rely on evidence sourced from various parties such as experts, witnesses, and law enforcement. However, this evidence, whether direct (like eyewitness accounts) or indirect (circumstantial), isn't always completely reliable. Forensic scientists must present theories based on this evidence in court to convince judges or juries. The prosecution aims to prove guilt, while the defense seeks to cast doubt on the prosecution's case rather than prove innocence. When presenting forensic evidence in court, both sides must persuade the judge or jury of its validity, which can be challenging since many court members lack scientific training. In cases lacking eyewitness testimony, reliance on circumstantial forensic evidence becomes crucial.

In a criminal trial, two aspects must be demonstrated: the criminal act (*Actus Reus*) and the intention behind it (*Mens Rea*). Forensic science is adept at determining the *Actus Reus* through physical, biological, and chemical evidence collected at crime scenes. However, it struggles to ascertain *Mens Rea* since it pertains to the mental state of the perpetrator, which cannot be observed directly in the physical world.

Forensic science encompasses various specialized units within a laboratory, each handling different aspects of scientific analysis. These units typically include a physical science unit, responsible for chemical and physical analyses like spectroscopy and microscopy, as well as examinations of drugs, materials such as glass and paint, explosives, soil, and other miscellaneous items. A biology unit may also exist to analyze biological samples like hair or botanical samples such as plants, with DNA analysis being a crucial capability. Firearms units handle investigations involving guns, bullets, and cartridge cases, while document examination units specialize in detecting forged documents through handwriting, print, paper, and ink analysis. Photography units are essential for recording and analyzing photographic evidence. Toxicology units analyze body fluids for toxins, and fingerprinting units provide expertise and technology for fingerprint analysis and visualization. Computer forensics is increasingly vital for retrieving deleted data and tracing electronic trails, while forensic entomology uses insect populations at crime scenes to gather information. Forensic anthropology involves analyzing skeletal remains for identification, especially in cases where only skeletons are found due to

decomposition. Odontology, the study of teeth, aids in identifying individuals by comparing dental records with teeth found at crime scenes.

Forensic scientists not only identify items but also classify them based on class characteristics, placing them in specific subgroups. Reconstruction in forensic science involves understanding the sequence of past events to narrate how an incident occurred, distinct from re-enactment, which recreates events to test hypotheses or understand unusual phenomena. Re-enactment may be used when events involve circumstances beyond typical experience or require testing of specific phenomena to ascertain what truly happened.

Atomic Structure & Spectroscopy

In forensic science, identifying objects often requires understanding their composition, including the presence of chemical elements. Spectroscopy and knowledge of atomic structure are fundamental for elemental analysis in forensic science. The question of when to conduct elemental analysis in forensic science is answered pragmatically: whenever it provides essential information for the investigation.

An element, as defined by the Anglo-Irish chemist Robert Boyle in the 17th century, is a substance that cannot be created or destroyed and cannot be broken down into simpler materials. This concept of elements being fundamental building blocks remains crucial in forensic science. However, it's important to note that while elements cannot be broken down by chemical means, they can be created or destroyed through nuclear processes in reactors.

When a human body is burned, its elements aren't destroyed but transformed into different forms. For example, carbon becomes carbon dioxide, hydrogen turns into water, and nitrogen may become nitrogen oxides. Even elements like gold or mercury from dental fillings remain after burning, albeit in altered states such as gold and mercury oxides. These elements, although not naturally part of the body, can still be detected afterward.

Consider the composition of objects like bullets. While most bullets are predominantly lead, none are entirely pure lead. They contain trace impurities, such as silver and antimony, alongside the main component. These trace impurities vary between batches of bullets, making them valuable for identifying the bullet's origin.

Elemental analysis involves determining both the bulk composition and detecting trace impurities present in small amounts. For forensic scientists, analyzing trace impurities often provides more useful information. Trace elements are measured in parts per million (ppm) or parts per billion (ppb), indicating the concentration of the element in the material.

Spectroscopy is essential for conducting analysis at minute levels. It operates on the principle that when light passes through a substance, it disperses into its constituent colors, forming a spectrum. This phenomenon is commonly observed in rainbows, where sunlight is refracted and separated into its component colors by water droplets in the air.

At times, forensic scientists must analyze extremely small objects, such as gunshot residue particles, which are typically around five micrometers in size—far too tiny to be seen with the naked eye. To visualize these objects, a Scanning Electron Microscope (SEM) is employed. Unlike conventional microscopes that use light, SEMs use electrons. In an SEM, an electron gun

emits a beam of electrons directed at the object being examined. These electrons interact with the object's surface, producing signals that are detected and used to generate an image, as depicted here. One of the advantages of SEM is that it is non-destructive.

Molecular Spectroscopy & Chromatography

An added challenge in analyzing compounds, unlike elements, is that the samples are often complex mixtures. Consider a blood or urine sample, which contains numerous compounds resulting from metabolic processes. Forensic scientists may need to sift through this intricate mixture to isolate just one or a few compounds of interest.

When addressing compound analysis, two main aspects come into play. Firstly, the separation of the mixture to pinpoint the desired compound(s), and secondly, the identification of the compound(s). Chromatography is the primary method used for separating mixtures, with various types available, all based on the same principle. One commonly used technique is Thin Layer Chromatography (TLC), which involves a TLC plate consisting of an inert backing material coated with a thin layer of absorbent material, typically silica or alumina.

Gas Chromatography (GC) surpasses Thin Layer Chromatography (TLC) in several aspects, primarily its heightened sensitivity and superior resolution. Unlike TLC, where the mobile phase is liquid, GC employs a gaseous carrier phase. The output of GC is represented by peaks on a graph, allowing for tentative identifications by comparing these peaks to known standards. However, just like with TLC, these identifications are not always definitive.

One technical difference between GC and TLC is the appearance of a solvent peak in GC chromatograms, which is disregarded during analysis. Additionally, GC offers quantifiability, as the amount of material present is proportional to the area under the peak. This capability enables both the separation of mixtures into their components and the measurement of their relative amounts.

Another important chromatographic technique in forensic science is High Performance Liquid Chromatography (HPLC). Similar to GC, HPLC measures the time it takes for each component to traverse the column, yielding sharp, well-resolved peaks. HPLC, like GC, can also be quantified based on the area under the peak, reflecting the component's quantity in the mixture. While GC equipment tends to be simpler and less expensive than HPLC, HPLC is advantageous for analyzing non-volatile and thermally unstable samples. Both methods are widely utilized in compound analysis within forensic science.

The different chromatography techniques serve the purpose of separating mixtures into individual components, offering initial insights into their composition through comparison with authentic samples. However, for conclusive identification suitable for court proceedings, additional techniques are necessary, such as spectroscopy and spectrometry.

Infrared Spectroscopy is particularly valuable for compound identification in forensic contexts. This technique is based on molecular vibrations, wherein molecules exhibit flexion, stretching, and wobbling motions. When irradiated with infrared light of the appropriate frequency,

molecules vibrate, and the absorbance of infrared light is monitored, forming the basis of infrared spectroscopy.

For forensic scientists, seeking to individualize compounds is crucial. In this region, vibrations and deformations specific to each compound create a molecular fingerprint. By comparing the infrared spectrum of an unknown compound to that of a standard sample, a strong identification of the compound can be achieved if they match in this region.

Mass Spectrometry is a significant technique for analyzing molecules, primarily by measuring their mass. To accomplish this, the sample is introduced into a mass spectrometer under high vacuum conditions, where it vaporizes. The molecules are then converted into positively charged ions, typically by bombarding them with electrons, before being accelerated by a high voltage and passed into a strong magnetic field. As the ions move through the magnetic field, they are deflected, with lighter molecules experiencing greater deflection. By measuring the degree of deflection, the mass of the molecule can be determined. For example, cocaine, with its molecular formula $C_{17}H_{21}NO_4$, has a calculated molecular weight of 303.

However, determining the molecular weight alone is insufficient for identification because many different arrangements of atoms can result in the same weight. These variations, known as isomers, complicate identification. Yet, Mass Spectrometry offers another crucial capability: analyzing fragmentation patterns. When molecules are bombarded by electrons, they often break into fragments, each with a lower weight than the original molecular ion. By analyzing the pattern of these fragments, unique to each compound, Mass Spectrometry can aid in identification.

Gas Chromatography-Mass Spectrometry (GC-MS) is a widely used technique that combines the separation capabilities of gas chromatography with the identification capabilities of mass spectrometry. In GC-MS, the output from the gas chromatograph directly feeds into the mass spectrometer, allowing for simultaneous separation and identification of components in a mixture.

Overall, chromatographic techniques, such as TLC, GC, and HPLC, are employed to separate complex mixtures into individual components, which are then identified using spectroscopic and spectrometry techniques like Infrared Spectroscopy and Mass Spectrometry. The combination of separation and identification into a single instrument, as seen in GC-MS, is particularly valuable in forensic science.

Time of Death

Determining the time of death is crucial in forensic investigations, particularly in cases of murder, where it aids in reconstructing the sequence of events and correlating suspect movements. Several techniques are employed for this purpose, beginning with assessing the cooling of the body, known as Algor Mortis, which occurs as the body's biochemical processes cease. However, the rate of cooling can be influenced by various factors, including body size, clothing, environmental conditions, and measurement location. Another phenomenon, Rigor Mortis, involves the stiffening of muscles due to ongoing chemical activity after death. Rigor Mortis appears first in the jaw, then spreads to the arms and legs, disappearing gradually over time. Expert assessment is essential in evaluating the state of Rigor Mortis accurately. Livor

Mortis results from the settling of red blood cells due to gravity after the heart stops beating. The position of the body influences the distribution of these blood cells, leading to distinct discoloration patterns that can provide additional clues about the circumstances of death.

While these methods rely on expert opinion, efforts are underway to develop more scientific approaches. Ocular potassium, for instance, involves measuring the potassium concentration in the eye's vitreous humor, which increases gradually after death due to diffusion from the rest of the body.

Circumstantial evidence, such as stomach contents or the status of personal items like watches and mobile phones, can also offer insights into the timing of death. For instance, examining mobile phone records can provide valuable data on the last use of the device, helping to establish a timeline of events leading up to the individual's death. Overall, while estimating the time of death remains a complex task influenced by numerous factors, advancements in technology and forensic techniques continue to enhance investigators' capabilities in this critical aspect of forensic science.

When dealing with a body that has undergone significant decomposition, experts often turn to the process of putrefaction for clues. Putrefaction occurs as microorganisms within the body continue to metabolize tissues, producing various chemicals, including gases, which give the body its characteristic odor. Other factors, such as temperature and drug use, can influence the rate of putrefaction, requiring careful consideration during examination.

During putrefaction, the body undergoes several changes. Discoloration and swelling occur as gases inflate the body, causing blisters and eventual skin tightness. Tissues break down, leading to liquefaction, making identification difficult as facial features become unrecognizable. Ultimately, in exposed environments, the body will decompose into a skeleton, a process influenced by environmental factors. However, in certain conditions, such as extreme dryness, mummification may occur, preserving the body's tissues. While accidental mummification can result from environmental circumstances, deliberate mummification, as seen in historical figures like Pharaoh Ramses II, Lenin, Chairman Mao, and Jeremy Bentham, involves intentional preservation methods, including chemical treatments. These examples highlight the varied outcomes of decomposition and preservation in different environmental and cultural contexts.

Forensic Entomology

An additional technique for bodies that are in this stage is to use what is called forensic entomology. Entomology is the study of insects. So, how can we use insects to determine the time of death? Well, our dead body is lying somewhere outdoors and it's starting to smell and it's detected very, very quickly by insects, and the females, of course, are always looking for a good place to lay their eggs, and there's nowhere better than a dead body. They can lay their eggs on the dead body; when the eggs hatch, their children come out, that's the maggots, then their children are assured of a really good supply of food, so that they can develop and produce the next generation of blowflies.

The life cycles of these creatures are usually fairly regular. So if you can find a maggot off the corpse, and you can estimate how old that maggot is, you've got a good estimate of the time of

death. But it has to be done right because each kind of insect has its own life cycle, with its own timing. The first thing you have to do is to know the species.

This is not so easy from a maggot. So, what is typically done is that the maggots are collected from the body, from the corpse, they are taken back to the laboratory, and they are reared to adulthood. When you have an adult, it is much easier to identify the species. When you identify the species, then of course, you know what the life cycle is. So when you take your maggot back to the laboratory and you rear it to adulthood, you also time how long it takes to reach the adult stage, and then you can back calculate, from the length of the life cycle, how long it had been on the body.

Skeletal Remains

Ultimately, the body is going to turn into a skeleton, and once it's a skeleton, it's very difficult to get good information because bones, after all, are bones. One test that can be done is to look at fluorescence.

Fresh bone will fluoresce under UV light. Old bones, maybe more than a hundred years old, will no longer fluoresce. And this is very useful because if bones turn out to be more than a hundred years old, then you can be pretty safe in assuming that anyone associated with that event is also going to be dead by this time.

You can do some analysis to get some more information. What we can do is use techniques of elemental analysis to measure the level of specific elements in the bone, and this can enlighten us a little more. This is typically called FUN analysis after three elements: fluorine, uranium, and nitrogen.

Now, fresh bone contains proteins. All proteins contain the element nitrogen. And in fresh bone, if you analyze it, you will find that bone is about 4% nitrogen by weight. However, let's suppose the body is in a grave or somewhere exposed. The proteins in those bones will gradually break down with time. But as the protein molecules break down, the nitrogen-containing compounds will diffuse out of the bone into the environment and be lost. Which means that the bone nitrogen content will decrease with age.

Now, let's turn to the upper two elements, fluorine and uranium. These two elements are not naturally present in bone, so the content of these two elements in fresh bone is 0%. There shouldn't be any fluorine or uranium in your bones. Now suppose those bones are buried in an area where the groundwater contains appreciable amounts of those elements, okay, they would be leaked from the local rocks. Then, the bones will absorb those elements from the environment. And that means that the concentration of fluorine and uranium will increase with time.

Now, you cannot use this as a kind of clock to determine the time of death because the absorption of these elements depends on their presence in the local environment. So if that body is in a place where the local geochemistry means there is no uranium, then of course, it will not absorb any uranium however long you leave it. But if it's in an area where there is uranium naturally present as a trace mineral in the rocks, then you will get uranium absorption. And if those two bones have been in the same place for the same length of time, the same number of centuries maybe, then they should have the same amount of uranium and fluorine.

Carbon-14 Dating

What if the human remains that have been found are really old? What technique can we use for dating them? The answer is a technique called carbon-14 dating.

Carbon is ubiquitous in living organisms; it's the basis of all life on Earth, and it comes in three forms. By far, the most common is carbon-12, this is the isotope carbon-12. So in carbon-12, the nucleus of the atom is made up of 6 protons and 6 neutrons, and it's a quite stable nucleus. A very small amount of carbon is carbon-13, and in carbon-13, the nucleus of the atom is made up of six protons and seven neutrons, and again, this is a perfectly stable atom. But there's a small amount of carbon-14, and in carbon-14, the nucleus is made up of six protons and eight neutrons, and this is an unstable nucleus. That means carbon-14 is radioactive. And if you have a sample of carbon which contains 12, 13, and 14, gradually, slowly over time, that carbon-14 will decay and disappear.

Radioactive decay occurs at a very fixed and regular rate, and for carbon-14, we know what the rate is and we measure that rate by the so-called half-life. A half-life is the time it takes for half the atoms in your sample to undergo radioactive decay. For carbon-14, that is just over 5000 years, so in 5000 years, half your carbon-14 will disappear. In another 5000 years you'll be down to a quarter, and another 5000 years down to an eighth, and so on and so on and so on.

How can we use this for dating? The key assumption is, the amount of carbon dioxide which is carbon-14 in the atmosphere is approximately constant. The carbon-14 is produced by the bombardment of the atmosphere by cosmic rays. As long as the bombardment by cosmic rays is constant, then the formation of carbon-14 is going to be constant. Now where does the carbon-14 go? It's present in the atmosphere in carbon dioxide. So whenever a plant absorbs some carbon dioxide from the air, it absorbs that amount of carbon-14, and that carbon-14 will then make its way through the food chain and end up in herbivorous animals, carnivorous animals, us, anything made of wood, etc. So we continually intake small amounts of carbon-14 from the diet, and of course, we give out carbon-14 when we breathe, which means it reaches a steady state. So in a living organism, the amount of carbon-14 is at a steady state during life, but of course when you die, all this stops. But the decay of carbon-14 through radioactive processes doesn't stop. So after death, the amount of carbon-14 drops with radioactive decay at a very nice and fixed mathematical rate.

So Willard Libby, who won the Nobel Prize in 1960, developed this as a dating method and he was able to calibrate it against historical objects, so we can have a great deal of confidence in this method. The limitation on carbon-14 is that the object you're studying must be more than a few centuries old, more than, say, 400 years old. And the reason for that is very simple because if something is only 100 or 200 years old, the amount of carbon-14 that has decayed is too small for us to measure accurately. So if something is 500 years old, then we can get a good measurement; if something is 10,000 years old, we can also get a good measurement. Future archaeologists are going to have a problem, and that is because the assumption of the carbon-14 in the atmosphere being constant has now broken down, and this is due to testing of atomic bombs in the atmosphere back in the 50s and the 60s. Those atomic bomb explosions produced all sorts of isotopes including carbon-14. And you can see on the graph here that prior to atom bomb testing, the carbon-14 level was nice and steady, and then it jumps up to a big spike

because of atom bomb testing, and is slowly dropping away, but it's going to take some time for it to get back to normal.

This method can help verify the authenticity of archaeological or historical artifacts, aiding in cases involving forged evidence. By dating bone or tissue samples, forensic scientists can confirm the age and legitimacy of artifacts. Carbon-14 dating also assists in comparing samples from different crime scenes, linking evidence chronologically and connecting different pieces of evidence to a single event or individual. In cold case investigations, where traditional methods may have failed, carbon-14 dating can provide new insights by dating previously unsolved cases, potentially leading to breakthroughs in solving crimes.

Blood

The next topic is blood and the forensic uses of blood samples, also known as forensic serology. In a violent crime, blood is likely to be spilled, whether it's on the victim, at the crime scene, or on the suspect. Fresh blood is easy to identify, but dried blood may require tests to confirm. Once blood is found and identified, the question becomes: is it human blood, and if so, whose? Blood can also provide clues about how a crime unfolded based on its pattern.

Blood is a complex substance, constituting about 8% of our body weight, primarily composed of plasma and various biochemicals. Blood serves as a transportation system for nutrients, waste, and gases like oxygen. It contains different types of blood cells with specific functions, such as red blood cells transporting oxygen and white blood cells facilitating immune responses.

Blood types are determined by antigens on the surface of red blood cells, which also influence transfusion compatibility. Despite common beliefs associating blood type with personality, its scientific significance lies in immunological reactions and forensic investigations. While blood type cannot definitively prove guilt, it can sometimes exonerate suspects. However, advancements in DNA technology have largely replaced blood typing in forensic investigations, as DNA provides more individualized identification.

Tests for Blood

There are several simple chemical tests we can use as presumptive tests for blood. One of the old ones is called the Benzidine test, but it's not commonly used anymore due to the toxicity of the chemicals involved. An excellent test that gives a nice color reaction is the Kastle-Meyer test using phenolphthalein, and another reliable reagent for testing for blood is Luminol. All of these tests are based on the chemical property of blood, which can catalyze the breakdown of hydrogen peroxide to water and oxidizing species.

Let's take a closer look at the Kastle-Meyer test. To perform this test, we need reduced phenolphthalein. Phenolphthalein in its acidic form is colorless, but in alkali, it turns into a bright pink color. When the alkaline form is reacted with zinc powder, it's reduced back to a colorless state. In the presence of an oxidizing agent, the reduced form is converted back into the bright pink form. Blood serves as an excellent catalyst for this reaction.

So, if we have a suspected blood stain, we treat it with reduced phenolphthalein and hydrogen peroxide. If we observe a color change to pink, then it's likely blood. However, this test is still only presumptive and doesn't definitively prove that it's blood, as other materials can also

catalyze this reaction. For example, potatoes contain powerful enzymes that can give a false positive in the Kastle-Meyer test.

If you've performed your Kastle-Meyer test or your Luminol test and determined that a stain found at the crime scene is indeed blood, there's still the question of whether it's human blood or belongs to some other animal. Antibody technology can help us determine the species of blood.

Here's how it works: A sample of human blood is injected into a rabbit. The rabbit's immune system reacts to the presence of the human blood by producing antibodies against its components. The serum from the rabbit's blood, containing these antibodies, can then be used as the test reagent. When the suspected human blood is treated with the rabbit serum, the antibodies in the serum will bind to components in the human blood, indicating an interaction.

This testing is typically done in a gel. The antigen (the suspected blood sample) and the antibody (the rabbit serum containing antibodies against human blood) are placed in separate wells in the gel. Using an electric field, they are induced to migrate towards each other. If there is an interaction between the antibody and the sample, when the two meet, a visible precipitin line forms in the gel. If there's no interaction, then no visible line appears, indicating that it's not human blood.

Blood Spatter Analysis

So, let's delve into blood spatter analysis and how it can help reconstruct a crime scene from the pattern of blood. Blood spatter analysis is the examination of the distribution of blood on a suspect or at a crime scene. Here are some typical blood patterns that might be found:

- **Transfer Stains:** These occur when blood is transferred from one surface to another. For instance, a partial bloody footprint or a handprint smeared on a wall.
- **Arterial Spurting:** When an artery is severed, blood spurts out in a characteristic pattern due to the variation in pressure as the heart beats. This results in a splash of blood on surfaces, followed by dribbling down.
- **Blood Drips:** This pattern occurs when someone with a small wound is standing around, resulting in a pattern of blood drips on surfaces.
- **Low, Medium, and High-Velocity Blood Spatter:** These patterns depend on the energy of the event causing the blood to fly. Low velocity spatter results from relatively low-energy events like a fight, while high velocity spatter is associated with gunshot wounds.

When analyzing blood spatter patterns, forensic scientists can determine various aspects of the crime, such as the direction and intensity of the force that caused the blood to spatter.

Consider the shape of a falling blood drop. If it hits the floor at an angle, it creates an elongated shape, and if it has enough energy, there may be a little splash in front of the main shape. The shape of blood droplets correlates with the angle at which they hit the surface. By analyzing multiple blood droplets in different places and plotting lines backward from them, forensic analysts can determine the point of convergence, which indicates the source of the blood.

This analysis is usually done in three dimensions to accurately reconstruct the crime scene. By plotting lines back from bloodstains on surfaces to the point of convergence, analysts can

determine the height above the floor at which the event generating the blood occurred. This information is crucial for understanding the sequence of events in the crime.

DNA

DNA technology has revolutionized forensic science, making Edmond Locard's statement, "Every contact leaves a trace," almost entirely true. With DNA technology, traces left behind at crime scenes can often be individualized to a particular person. One significant area where DNA technology has made a profound impact is in sex crimes. Previously, in cases of sexual assault or rape, there might have only been the victim and the perpetrator, with the latter often being the only survivor in cases of rape-murder. However, DNA technology enables forensic scientists to identify perpetrators even in the absence of witnesses.

Now, let's delve into what DNA is. The genetic material in humans is housed within the nucleus of most cells, except for red blood cells. Inside the nucleus, there are complete instructions to create a human being encoded in DNA, a molecule that weighs only about seven picograms. This incredible amount of information dwarfs the memory capacity of even the most advanced laptops. However, DNA is not found directly at crime scenes. Instead, biological materials containing DNA, such as blood, semen, saliva, skin cells, hair, and body parts, are collected and taken to the laboratory for DNA extraction.

Human DNA is organized into 23 pairs of chromosomes, totaling 46 chromosomes. Twenty-two pairs are normal chromosomes containing genetic information, while the remaining two determine sex: XY for males and XX for females. Forensic scientists can determine gender by analyzing these chromosomes, which exhibit distinctive patterns when dyed and viewed under a microscope. Chromosomes consist of DNA wrapped around protein molecules called histones, which help maintain its structure. Within chromosomes, genes are located in the coding region, which holds the instructions for creating a human being. Surprisingly, only a fraction of DNA, known as the coding region, contains genes, while the rest, comprising about 90 to 95% of DNA, is non-coding. This non-coding region, often referred to as junk DNA, serves an unknown purpose.

Understanding DNA's structure and organization is crucial for forensic scientists in analyzing DNA evidence and identifying individuals involved in criminal cases.

DNA Profiling

DNA profiling, also known as DNA typing or DNA fingerprinting, was pioneered by Professor Sir Alec Jeffreys at the University of Leicester several decades ago. Initially, Jeffreys used a technique called Restriction Fragment Length Polymorphism (RFLP), which has since been replaced by more advanced methods. This groundbreaking technique allows for the analysis of an individual's DNA profile, primarily using junk DNA, or non-coding DNA, which exhibits high variability between individuals.

The choice of junk DNA for profiling is due to the fact that the coding region, which contains genes responsible for traits like hair color and eye color, is nearly identical among humans. Therefore, using coding DNA for profiling would not provide sufficient individualization. In contrast, junk DNA, which constitutes about 90 to 95% of the genome, exhibits significant variation between individuals and is therefore ideal for DNA profiling.

A typical DNA profiling procedure involves several steps. First, the DNA molecule is fragmented using special enzymes called restriction enzymes. These fragments are then separated using a technique called electrophoresis, which is performed on a gel. The separated fragments are then transferred onto a nylon membrane for further analysis. To identify specific DNA fragments of interest, radioactive DNA probes, complementary to the target sequences, are used. When exposed to these probes, the target DNA fragments bind selectively, allowing them to be visualized using X-ray film.

Electrophoresis, the key technique in DNA profiling, involves applying an electrical potential across a gel matrix containing DNA fragments. The fragments migrate through the gel at different speeds, depending on their size and charge, resulting in their separation. Gel electrophoresis can be likened to thin layer chromatography (TLC), while capillary gel electrophoresis, a more advanced method, is analogous to high-performance liquid chromatography (HPLC).

By employing DNA profiling techniques, forensic scientists can analyze biological materials collected from crime scenes and generate individualized DNA profiles. These profiles can be compared to DNA samples from suspects or databases to identify perpetrators and aid in criminal investigations.

Polymerase Chain Reaction

Alec Jeffreys initially required relatively large samples for analysis. However, many traces found at crime scenes are extremely small and cannot be analyzed using traditional techniques alone. The advent of the polymerase chain reaction (PCR) has revolutionized forensic DNA analysis by allowing for the amplification of minute DNA samples.

Here's how PCR works. The double-stranded DNA sample is heated to around 94 degrees Celsius, breaking the hydrogen bonds between the two strands and causing them to separate. Short DNA primers, which are complementary to the ends of the DNA strands, are added at a lower temperature (around 60 degrees Celsius), allowing them to bind to their complementary sequences on the single-stranded DNA. At around 72 degrees Celsius, DNA polymerase enzymes are introduced along with the four nucleotides (A, T, C, G). The DNA polymerase recognizes the primers and synthesizes new DNA strands complementary to the original template strands.

By cycling through these steps multiple times, each cycle results in a doubling of the DNA target sequence. This exponential amplification allows for the generation of millions of copies of the target DNA from even minute amounts of starting material. PCR is typically performed in a thermal cycler machine, which automates the process by precisely controlling temperature changes during each cycle.

The result of PCR is a significant increase in the amount of DNA, making it possible to analyze very small traces, even those consisting of just a few cells. This capability, known as low copy number (LCN) DNA analysis, has greatly expanded the range of forensic samples that can be analyzed. For example, PCR can be used to analyze DNA from saliva left on a cigarette butt, providing valuable forensic evidence from trace biological samples.

Short Tandem Repeats

Short Tandem Repeats (STRs) are the fragments of DNA commonly used in forensic DNA profiling. These regions consist of repeating sequences of three, four, five, six, or seven base pairs. The variability between individuals lies in the number of times these sequences are repeated. For example, the THO1 STR might consist of the sequence AATG repeated multiple times. Each person inherits one copy of the STR from each parent, resulting in a combination of repeat numbers.

By analyzing multiple STR loci simultaneously through a process called multiplexing, forensic scientists can increase the accuracy and reliability of DNA profiling. Different databases may utilize varying numbers of STR loci for analysis. For instance, the U.S. national database typically uses 13 loci, while the U.K. national database uses 10. The more loci analyzed, the higher the probability of individualization.

During gel electrophoresis, the different STR loci produce distinct peaks, with each pair of peaks representing the alleles inherited from the mother and the father. In cases where both alleles at a particular locus are the same, it suggests a homozygous genotype, where the individual inherited identical alleles from both parents.

DNA analysis is frequently employed in paternity and maternity testing. By comparing the STR profiles of the child with those of the alleged parents, forensic scientists can determine biological relationships. A complete match between the child's alleles and those of both parents confirms paternity or maternity. In cases of siblingship testing, the analysis involves comparing the profiles of the siblings with those of the parents to establish whether they share the same biological parents.

Identical twins present a unique challenge in forensic DNA analysis. Because they originate from a single fertilized egg, identical twins possess virtually identical DNA profiles. However, advancements in DNA sequencing technology may eventually enable the detection of subtle genetic differences between identical twins arising from spontaneous mutations during cell replication.

Mitochondrial DNA

Mitochondrial DNA (mtDNA) is distinct from nuclear DNA and is found within the mitochondria, the cell's energy-producing organelles. Unlike nuclear DNA, which is organized as long linear strands, mtDNA is circular in structure and contains 16,569 bases organized into 37 genes, including non-coding regions that exhibit variability among individuals.

One notable characteristic of mtDNA is its high copy number per cell compared to nuclear DNA. While there is only one copy of nuclear DNA per cell, there are multiple copies of mtDNA. This high copy number and the circular structure of mtDNA make it more robust and resistant to sample degradation. As a result, mtDNA can be extracted and analyzed from biological samples that are old or have been exposed to harsh conditions, where nuclear DNA may be more degraded and difficult to analyze.

An important distinction between mtDNA and nuclear DNA is the mode of inheritance. Mitochondrial DNA is inherited exclusively from the mother, not the father. This means that all individuals within a maternal lineage share the same mtDNA. As a result, mtDNA analysis is not useful for individualizing samples but can provide information about maternal lineage.

Furthermore, mtDNA undergoes faster mutation rates from generation to generation compared to nuclear DNA. This is partly due to the lower fidelity of the DNA replication process in mitochondria, which lacks the robust proofreading mechanisms present in the nucleus. As a result, mtDNA sequences can accumulate mutations more rapidly over time, which can be useful in evolutionary studies and population genetics.

Fingerprinting

Fingerprints are indeed an enduring method of identification, with roots tracing back centuries. While DNA has become the gold standard in forensic science, fingerprints remain a vital tool in police work, offering a unique and reliable means of identifying individuals. The history of fingerprinting is rich and varied, with examples of its use dating back to ancient China, where they may have served as signatures. William Herschel's innovative use of fingerprints in colonial India for signing documents further demonstrates their early recognition as a form of identification. Henry Faulds' pioneering work in the late 19th century, culminating in his proposal of fingerprints for criminal identification, laid the foundation for their widespread adoption in forensic science. Though initially met with skepticism by law enforcement, fingerprints eventually gained acceptance and have since become a cornerstone of criminal investigations. Even in fiction, authors like Mark Twain recognized the dramatic potential of fingerprints, incorporating them into stories such as "The Tragedy of Pudd'nhead Wilson."

Friction ridges, those intricate patterns on our skin, serve a crucial purpose in our ability to grip objects securely. These ridges, formed by glands just beneath the skin's surface, excrete a mixture of natural chemicals, including salts, fats, amino acids, proteins, and water. When deposited onto surfaces, this mixture creates fingerprints, unique to each individual.

Comparing fingerprints to DNA reveals some key differences. While DNA technology allows for statistical calculations indicating the extreme rarity of identical DNA profiles, fingerprinting relies more on empirical observation. Over more than a century of use, no two individuals have been found to have identical fingerprints, not even identical twins. This underscores the first principle of fingerprinting: the uniqueness of each individual's prints, including those of identical twins.

Another fundamental principle is that fingerprints remain unchanged throughout a person's lifetime. They form before birth and persist after death, enduring as long as the skin remains intact. This permanence renders fingerprints invaluable for forensic identification purposes.

Fingerprint patterns can be classified based on both small features and overall patterns. Early work by Galton laid the groundwork for this classification system, which includes terms inspired by geography and transportation. These classifications enable fingerprint examiners to describe and categorize fingerprints systematically, aiding in their analysis and comparison.

Latent Fingerprints

Latent fingerprints, often invisible to the naked eye, require special techniques for visualization, typically involving chemical treatments. Understanding the composition of fingerprints—metal salts, amino acids and proteins, fats, and water—provides insight into these methods.

The simplest method involves dusting with powder, where the powder adheres to the fingerprint, making it visible. Different powders, such as carbon black for light surfaces or aluminum for dark ones, create contrast with the background. Once dusted, the fingerprint can be lifted onto tape, photographed, and preserved.

Chemical reagents offer another approach. Ninhydrin, for example, reacts with amino acids, rendering latent fingerprints visible. Other chemicals like iodine or silver nitrate can also be used. Physics-based methods, such as laser fluorescence, either alone or combined with a dye, provide additional visualization options.

One particularly effective method is fuming with cyanoacrylate, commonly known as super glue. When heated, cyanoacrylate forms a white residue on the fingerprint, making it visible. This method, despite its high-tech sounding name, is a straightforward and widely used technique in forensic science.

Firearms

The study of firearms in forensic science is typically divided into three main areas:

- **Internal Ballistics:** This aspect focuses on what occurs while the projectile is still inside the barrel of the firearm. It involves the study of factors such as the pressure generated by the propellant, the speed and acceleration of the projectile, and the behavior of the gunpowder or other propellant materials.
- **External Ballistics:** External ballistics deals with the projectile's behavior once it leaves the gun and is in flight towards the target. Factors such as gravity, air resistance, wind speed and direction, and the shape and weight of the projectile all influence its trajectory and accuracy.
- **Terminal Ballistics:** Terminal ballistics pertains to what happens when the projectile strikes the target. This includes the study of how the projectile penetrates the target, the resulting damage to tissues or objects, and any fragmentation or deformation of the projectile upon impact.

Understanding these aspects of firearms helps forensic scientists analyze evidence such as bullet trajectories, gunshot wounds, and the effects of different types of firearms and ammunition on targets.

Advancements in firearms forensics, such as the development of the breech loader and the invention of the cartridge, have revolutionized the efficiency and functionality of firearms. The breech loader, allowing loading from the front of the breech, has notably increased efficiency and firing rates. Meanwhile, the cartridge, comprising both the projectile (bullet) and the propellant (energy source), has streamlined the loading process, replacing the cumbersome task of separately loading the explosive charge and projectile in earlier firearms.

Cartridges are composed of various integral components: the bullet, which is the projectile aimed towards the target; the propellant, which supplies energy to the bullet, often composed of materials like black powder or nitrocellulose; the cartridge case, housing both the propellant and bullet; and the primer, a shock-sensitive compound situated in the cartridge's base, igniting upon mechanical shock to initiate the firing process.

The firing mechanism of a firearm involves pulling the trigger, releasing the firing pin to strike the cartridge's base, thereby causing the primer to detonate. This ignition sets off the propellant, leading to the firing of the gun. Forensically, the impression left by the firing pin on the cartridge base serves as valuable evidence, as each firing pin leaves a unique mark, aiding in firearm identification in criminal investigations.

Additionally, rifling, the process of cutting grooves along the barrel to impart spin to the projectile upon firing, enhances accuracy and stability. The characteristic scratch marks, known as striations, left by the grooves and lands in rifled barrels on the bullet serve as identifying features.

Comparison microscope analysis plays a pivotal role in forensic investigations, where test bullets fired from a suspect gun are compared with bullets recovered from the crime scene. Matching striations or firing pin impressions provide crucial evidence linking the firearm to the crime, assisting investigators in establishing connections between firearms, bullets, and crime scenes, ultimately aiding in the identification and apprehension of suspects.

Gunshot residue (GSR) serves as a vital forensic clue present near a discharged firearm. Composed of particles resulting from the combustion of propellant and primer components within the gun, GSR contains residual substances including nitrites, nitrates, and metal salts like lead, barium, and antimony. Detecting and analyzing GSR involves several methods. The Griess Test identifies nitrite residues, though its specificity is limited and may yield false positives from other nitrite sources such as fertilizers, urine, cosmetics, or tobacco. Another approach, the Sodium Rhodizonate Test, reacts with metal compounds in GSR, notably lead, barium, and antimony salts, prompting observable color changes. However, sole reliance on color tests may introduce inaccuracies. Modern techniques, exemplified by Scanning Electron Microscopy with Energy Dispersive X-ray Analysis (SEM-EDX), offer enhanced precision. SEM enables visualization of GSR particles, while EDX precisely identifies the elemental composition, furnishing conclusive evidence in forensic investigations.

Narcotics

In discussing illegal drugs, they can be categorized into three types:

- **Natural:** These substances are primarily derived from plants or other organisms. They have been utilized since ancient times for medicinal or religious purposes. Examples include morphine from the opium poppy, which has a long history of medicinal use, and hallucinogenic plants used by traditional healers or shamans for spiritual experiences.
- **Semi-Synthetic:** With advancements in modern chemistry, natural substances can be chemically modified to create new compounds with altered properties. These modified natural compounds are referred to as semi-synthetic. An example of a legal semi-synthetic compound is aspirin, derived from a chemical naturally found in willow trees.
- **Purely Synthetic:** These compounds are entirely artificial and are typically developed by the pharmaceutical industry for medical purposes. However, some synthetic compounds have been misused and abused, leading to addiction and other harmful effects. This category may also include the abuse of solvents like paint and glue.

Regardless of their origin, these chemicals have the potential to cause physical and psychological addiction, leading to substance abuse disorders.

Sometimes, determining whether these drugs originate from a single source or multiple sources may be crucial. So, how can forensic science help in this investigation? Synthetic drugs are created through chemical transformations starting from simpler molecules, often acquired commercially. This process usually involves multiple steps, forming intermediate compounds before reaching the final drug product. This synthesis isn't limited to clandestine labs; pharmaceutical companies also use similar methods to produce legal drugs.

While pharmaceutical drug synthesis is tightly regulated by agencies like the FDA, clandestine labs operate without oversight. In legal production, chemists prioritize purity, ensuring stringent quality control measures. However, clandestine labs lack such regulation and rarely perform quality checks on their products. Forensic scientists analyze drug samples not by examining the drug molecule itself, but by studying the impurities present. Each synthesis method introduces unique impurities, creating distinct impurity profiles. By comparing these profiles across samples, scientists can determine whether the drugs share a common source. This approach allows law enforcement to identify patterns in drug distribution networks, aiding efforts to disrupt illegal drug trafficking operations.

Forensic scientists may employ a range of analytical techniques, including chromatography and spectroscopy, to detect impurities in drug samples. Methods like GC (gas chromatography), HPLC (high-performance liquid chromatography), mass spectrometry, and IR (infrared spectroscopy) are commonly used. These techniques help identify and quantify molecules present in the sample.

Toxicology

Toxicology, the study of poisons and poisoning, encompasses various scenarios where individuals are exposed to harmful substances. Accidental poisoning can occur when someone ingests a harmful substance unintentionally. Environmental poisoning happens when individuals are exposed to toxic chemicals in their surroundings, like arsenic-contaminated groundwater. Occupational poisoning affects workers who are exposed to hazardous substances without adequate protection.

Furthermore, poisoning can be deliberate, historically used for suicide or murder. Poisoning as a means of homicide has a long history, but its prevalence has diminished over time. This decline is largely attributed to stricter regulations governing access to toxic substances. In the past, potent poisons could be easily acquired from pharmacies, but today, stringent controls limit their availability, reducing the incidence of poison-related crimes.

Toxins represent a distinct category of poisons, consisting of natural substances produced by living organisms. Similar to other poisons, toxins target vital biochemical processes within the body, causing harmful effects. These substances are often synthesized by various organisms, including plants, animals, fungi, and bacteria, as a defense mechanism or as a means to incapacitate prey or deter predators. Due to their biological origins, toxins can exhibit diverse chemical structures and mechanisms of action, posing significant challenges in their detection and treatment.

Paracelsus is regarded as the founder of toxicology. He lived during the first half of the 16th century and was a pivotal figure in shaping the scientific understanding of poisoning. Paracelsus introduced a crucial concept that laid the foundation for toxicology: "The dose makes the poison." This principle emphasizes that any substance, including those considered poisonous, can also have medicinal properties depending on the dosage. Paracelsus's insights challenged the prevailing beliefs of his time and paved the way for modern toxicological studies. His contributions underscore the importance of considering dosage levels when assessing the effects of substances on health.

The LD50, or lethal dose for 50%, is a measure of toxicity used to determine the amount of a substance required to cause death in half of the population of a specific test animal group. For example, if a group of mice is administered a poison and half of them die, the dosage given is considered the LD50. However, the accuracy of LD50 measurements can be limited. Factors such as the method of administration (e.g., oral ingestion versus injection) and the species of test animals used (e.g., mice, rats) can influence LD50 values. While LD50 values obtained from animal tests are often extrapolated to estimate toxicity in humans, this assumption may not always be accurate due to species differences in physiological responses to toxins.

LD50 values are typically expressed in units of milligrams per kilogram or micrograms per kilogram, representing the dose of a substance per unit body weight. This measurement standardizes the dosage across different species, accounting for variations in size. For example, mice require smaller doses than humans due to their smaller size. In addition to size, age also influences susceptibility to poisoning. Children and older adults are generally more vulnerable due to differences in metabolism. Liver health is another crucial factor, as individuals with liver disease may be more susceptible to toxins due to impaired metabolic function. Furthermore, an individual's tolerance to certain poisons can develop over time, affecting their response to repeated exposure. This historical context is important in understanding how the body reacts to toxins.

Glosary

Absorption spectroscopy. Measurement technique assessing radiation absorption in relation to frequency or wavelength.

Actus Reus. Criminal act leading to legal liability.

Algor Mortis. Body temperature decreases post mortem.

Alkaloid. Naturally occurring nitrogen-containing chemical compounds.

Ammunition. Loaded cartridges comprising case, propellant, and projectile.

Analog. Compounds sharing similar molecular structures.

Anthrax. Acute illness instigated by *Bacillus anthracis* bacteria.

Antibodies. Immunoglobulins countering specific antigens.

Antigens. Proteins on red blood cell surfaces eliciting immune responses.

Asphyxiation. Oxygen deprivation leading to suffocation.

Autosomal chromosomes. Typical genetic information carriers.

Back spatter. Blood propelled in reverse towards energy source surfaces.

Ballistics. Study of moving projectiles.

Barium meal. Ingested barium sulfate aiding soft tissue visibility in X-rays.

Base pairs. Complementary nucleic acid bases (e.g., adenine-thymine).

Blood spatter analysis. Utilization of blood patterns to reconstruct crime events.

Blood typing. Determination of blood group via antigen presence.

Botulism. Paralytic illness caused by *Clostridium botulinum* bacteria.

Breathalyzers. Devices measuring breath alcohol levels, utilizing chromium chemistry.

Breech loading firearms. Firearms loading cartridges from rear barrel chambers.

Bulk composition. Predominant material makeup.

Bullet. Projectile expelled from firearm.

Cartridge. Ammunition unit with case, primer, powder, and bullet.

Central Nervous System (CNS). Brain and spinal cord assembly.

Chromatography. Method for mixture component separation.

Circumstantial evidence. Indirect evidence implying factual occurrences.

Compound. Pure substance with multiple elements.

Co-polymer. Long-chain molecule of multiple monomers.

Derivatization. Chemical transformation technique.

Electromagnetic radiation. Radiation spectrum including visible light, gamma rays, etc.

Element. Pure substance consisting of single atom type.

Emission spectroscopy. Measurement of emitted electromagnetic radiation during electron energy level transitions.

Energy Dispersive X-ray Spectroscopy (EDX). Analytical technique for sample chemical characterization.

Energy level. Quantum state system energy.

Entomology. Study of insects.

Erythrocytes. Red blood cells without nuclei.

External ballistics. Bullet flight dynamics.

Facial reconstruction. Recreation of facial appearance from skeletal remains.

Firing pin. Mechanism part igniting cartridge primer.

Firing pin impression. Imprint on cartridge from firing pin strike.

Fluorescence. Substance emission upon external radiation exposure.

Forensic odontology. Dental evidence handling in legal contexts.

Friction ridges. Raised epidermal sections for object gripping, forming fingerprint patterns.

Gamma ray. Nuclear atom emission by electromagnetic radiation.

Griess Test. Nitrite detection chemical test.

Gun Shot Residue (GSR). Primer and charge explosion residues.

Hair cortex. Middle layer between hair cuticle and medulla, containing majority of hair pigments.

Hair cuticle. Outermost layer of hair shaft, composed of dead cells.

Hair follicle. Structure from which hair grows, housing hair roots.

Hair medulla. Innermost portion of hair shaft, exhibiting species-specific width and pattern.

Hair root. Base of hair within the follicle.

Hallucination. Perception of nonexistent objects due to neural communication disruption.

Hallucinogen. Substance inducing hallucinations.

High velocity spatter. Blood pattern from energetic events like gunshot.

Homologous chromosomes. Chromosome pair of similar size and length, one from each parent.

Human genome. Entire genetic material of a human including DNA, genes, and chromosomes.

Immunoassay. Method for substance identification based on antibody binding.

Individualization. Process of marking something as unique.

Infrared (IR) spectroscopy. Analysis of molecular interaction with infrared light.

Internal ballistics. Processes occurring within a gun barrel.

Kastle Meyer Test. Blood detection test using phenolphthalein.

Lands and grooves. Rifling features in barrel; lands are raised portions between grooves.

Latent prints. Hidden fingerprint requiring chemical visualization.

Lethal dose for 50 percent (LD50). Amount of substance to kill half the test population.

Ligature strangulation. Strangulation without body suspension using cord-like objects.

Livor Mortis. Blood pooling in lower body post-mortem causing skin discoloration.

Low velocity spatter. Blood pattern from low-energy events like dripping wounds.

Manifesto. Public declaration of aims or intentions.

Mass Spectrometry (MS). Instrumental technique for identifying substance constituents by ion separation based on mass and charge differences.

Medium velocity spatter. Blood pattern from moderately energetic events like baseball bat beatings.

Mens Rea. Guilty intention or state of mind.

Mitochondrial DNA (mtDNA). Circular DNA in mitochondria, passed down the maternal line.

Mobile phase. Fluid transporting mixture through stationary phase.

Molecule. Atoms bonded together chemically.

Monomer. Basic unit forming polymers, capable of linking to form chains.

Mummification. Preservation of a body as a mummy through embalming or drying.

Muzzle loading firearms. Guns loaded from the front end.

Neurons. Specialized cells transmitting nerve signals.

Neurotransmission. Nerve impulse transmission across synapses.

Neurotransmitter. Chemical facilitating nerve impulse transmission (e.g., acetylcholine).

Neutron Activation Analysis (NAA). Method determining material elemental composition via neutron bombardment.

Nucleotides. DNA basic units with sugar, phosphate, and base.

Part per billion (ppb). One in a billion ratio.

Part per million (ppm). One in a million ratio.

Poisons. Substances fatal or harmful to health.

Polymer. Long chain molecule comprising repeating units.

Polypeptide. Amino acid chain connected by peptide bonds.

Precipitin Test. Criminological blood stain source determination via antibody technology.

Precursor chemicals. Compounds used as starting materials in drug synthesis.

Primer. Shock-sensitive compounds igniting upon impact.

Propellant. Chemical providing energy when ignited.

Proteins. Large molecules composed of amino acid chains.

Putrefaction. Anaerobic organic matter decomposition by bacteria and fungi, emitting foul odors.

Radioactive. Emitting radiation due to atomic instability.

Radioactive decay. Spontaneous atomic nucleus transformation with radiation emission.

Reconstruction. Establishing event sequences through evidence study and interpretation.

Re-enactment. Repetition or acting out of past events.

Relative abundance. Element quantity in comparison to others in an environment.

Retention Factor (Rf). Ratio describing spot position in chromatography.

Retention Time. Time for solute travel through columns.

Rifling. Gun barrel spiral groove cutting; lands are metal between grooves.

Rigor Mortis. Post-mortem muscle stiffening due to chemical changes.

Route of administration. Path through which poison enters the body.

Scanning Electron Microscope (SEM). Electron microscope producing sample images via electron beam scanning.

Semi-synthetic drugs. Drugs chemically altered from natural counterparts.

Short tandem repeats (STR). Repeating DNA sequences of three to seven base pairs.

Skin graft procedure. Surgical transfer of skin from one body part to another.

Sodium Rhodizonate Test. Chemical test for lead particulates or vapors near bullet holes.

Species. Group of similar living organisms.

Stationary phase. Fixed chromatography layer interacting with analytes (e.g., silica in TLC).

Stereochemical. Molecule's three-dimensional structure difference.

Striations. Bullet groove-induced scratch marks.

Synthetic drugs. Substances solely chemically synthesized.

Terminal ballistics. Projectile impact on target.

Toxicology. Study of poisons and their effects.

Trace impurities. Small portion composition of an object.

Visible prints. Fingerprints discernible to the naked eye.

Xenobiotics. Foreign chemicals not naturally produced by an organism.